

Problem Set 9—Suggested Answer

We have $Y_1, \dots, Y_N$ iid with pdf

$$f(y|\mu) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}(y - \mu)^2\right),$$

i.e. $Y_i \sim N(\mu, 1)$.

(i) Log likelihood function.

The likelihood is the joint density evaluated at the observed data, treated as a function of the parameter:

$$L(\mu) = \prod_{i=1}^N f(y_i|\mu) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}(y_i - \mu)^2\right) = \left(\frac{1}{\sqrt{2\pi}}\right)^N \exp\left(-\frac{1}{2} \sum_{i=1}^N (y_i - \mu)^2\right).$$

Taking logarithms (a strictly increasing transformation that preserves the maximiser):

$$\ln L(\mu) = -\frac{N}{2} \ln(2\pi) - \frac{1}{2} \sum_{i=1}^N (y_i - \mu)^2.$$

(ii) MLE and unbiasedness.

Differentiate $\ln L(\mu)$ with respect to μ :

$$\frac{\partial \ln L}{\partial \mu}(\mu) = \sum_{i=1}^N (y_i - \mu).$$

Setting this equal to zero and solving:

$$\sum_{i=1}^N y_i - N\mu = 0 \implies \hat{\mu}^{\text{ML}} = \frac{1}{N} \sum_{i=1}^N y_i = \bar{Y}.$$

The second derivative is $-N < 0$, confirming this is a maximum.

Unbiasedness. Taking expectations:

$$\text{E}(\hat{\mu}^{\text{ML}}) = \text{E}(\bar{Y}) = \frac{1}{N} \sum_{i=1}^N \text{E}(Y_i) = \frac{1}{N} \cdot N\mu = \mu.$$

Hence $\hat{\mu}^{\text{ML}}$ is unbiased for μ .

(iii) **Variance of the MLE.**

Since the Y_i are iid with $\text{Var}(Y_i) = 1$:

$$\text{Var}(\hat{\mu}^{\text{ML}}) = \text{Var}(\bar{Y}) = \frac{1}{N^2} \sum_{i=1}^N \text{Var}(Y_i) = \frac{1}{N^2} \cdot N \cdot 1 = \frac{1}{N}.$$

(iv) **Score function.**

The score function for a single observation is defined as the derivative of the log density with respect to the parameter:

$$S(y|\mu) := \frac{\partial \ln f(y|\mu)}{\partial \mu}.$$

From the log density

$$\ln f(y|\mu) = -\frac{1}{2} \ln(2\pi) - \frac{1}{2}(y - \mu)^2,$$

we obtain

$$S(y|\mu) = \frac{\partial}{\partial \mu} \left[-\frac{1}{2}(y - \mu)^2 \right] = (y - \mu).$$

(v) **Fisher information and the Cramér–Rao lower bound.**

The Fisher information for a single observation is

$$I(\mu) := \text{E}(S(Y|\mu)^2) = \text{E}((Y - \mu)^2) = \text{Var}(Y) = 1.$$

For a sample of N iid observations, the **Cramér–Rao lower bound (CRB)** for any unbiased estimator $T(Y_1, \dots, Y_N)$ of μ is

$$\text{Var}(T) \geq \frac{1}{N I(\mu)} = \frac{1}{N}.$$

No unbiased estimator can have variance smaller than $1/N$.

(vi) **Confirming the information equality.**

The information equality states

$$I(\mu) = \text{E}(S(Y|\mu)^2) = -\text{E}\left(\frac{\partial S(Y|\mu)}{\partial \mu}\right).$$

We already found $I(\mu) = \text{E}(S(Y|\mu)^2) = 1$. For the right-hand side:

$$\frac{\partial S(y|\mu)}{\partial \mu} = \frac{\partial}{\partial \mu}(y - \mu) = -1,$$

so

$$-\text{E}\left(\frac{\partial S(Y|\mu)}{\partial \mu}\right) = -(-1) = 1.$$

Both sides equal 1, confirming the information equality. ✓

(vii) **Does $\hat{\mu}^{\text{ML}}$ attain the CRB?**

From part (ii), $\hat{\mu}^{\text{ML}}$ is unbiased. From part (iii), $\text{Var}(\hat{\mu}^{\text{ML}}) = 1/N$, which equals the CRB from part (v). Therefore $\hat{\mu}^{\text{ML}}$ *does* attain the Cramér–Rao lower bound and is the minimum variance unbiased estimator for μ .

(viii) **Decomposition of the average score.**

The theorem on minimum variance unbiased estimators states that if an unbiased estimator T attains the CRB, then the average score function can be written as

$$\frac{1}{N} \sum_{i=1}^N S(y_i|\mu) = a(\mu)(T(y_1, \dots, y_N) - \mu)$$

for some function $a(\mu)$.

Computing the average score directly:

$$\frac{1}{N} \sum_{i=1}^N S(y_i|\mu) = \frac{1}{N} \sum_{i=1}^N (y_i - \mu) = \bar{Y} - \mu = \hat{\mu}^{\text{ML}} - \mu.$$

Comparing with $a(\mu)(\hat{\mu}^{\text{ML}} - \mu)$, we immediately read off

$$\boxed{a(\mu) = 1.}$$

(ix) **Asymptotic distribution of $\hat{\mu}^{\text{ML}}$.**

By the general asymptotic normality result for the MLE,

$$\sqrt{N}(\hat{\mu}^{\text{ML}} - \mu) \xrightarrow{d} \text{N}\left(0, \frac{1}{I(\mu)}\right) = \text{N}(0, 1).$$

In this model the asymptotic variance equals the exact variance $1/N$ (scaled by N), which is not surprising given that $\hat{\mu}^{\text{ML}} = \bar{Y}$ and Y_i are exactly normal.